# Hikmet Burak Özcan, Ph.D.

**Research Assistant**

Department of Mathematics, İzmir Institute of Technology, İzmir, Türkiye

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## Profile

I am a mathematician working in harmonic analysis. My main topic is the regularity of discrete Hardy–Littlewood maximal operators. I also work on uniform distribution and number theory. I completed my Ph.D. at İzmir Institute of Technology in 2026, where I have worked as a teaching assistant for undergraduate mathematics courses in English since 2019. In my teaching I focus on clear explanations, guided problem solving and carefully written course materials.

## Research Interests

Harmonic analysis and maximal operators · Regularity of maximal functions · Uniform distribution of sequences · Algebraic number theory · Mathematical foundations of machine learning and data science

## Education

2020 – 2026 **Ph.D. in Mathematics**, İzmir Institute of Technology
Thesis: *On Regularity of Hardy–Littlewood Maximal Operators*
Supervisor: Asst. Prof. Faruk Temur

2018 – 2020 **M.Sc. in Mathematics**, Dokuz Eylül University
Thesis: *Prime Ideal Theorem on Number Fields*
Supervisor: Assoc. Prof. Haydar Göral

2011 – 2016 **B.Sc. in Mathematics**, İzmir University of Economics
GPA 3.57/4.00, graduated first in the class

## Academic Position

May 2019 – present **Research and Teaching Assistant**, Department of Mathematics, İzmir Institute of Technology

- Research in harmonic analysis, uniform distribution and number theory.
- Teaching assistant for undergraduate courses and TA coordinator of MATH 141 (see Teaching).
- Weekly tutorials for core courses, exam preparation, proctoring and grading.
- Coordination of the department seminar series; maintenance of the department website.

## Teaching

Fall 2026–2027 **MATH 141** Basic Calculus I – teaching assistant (TA) and TA coordinator
**MATH 457** Methods of Mathematical Economics – teaching assistant (TA)

Spring 2025–2026 **MATH 332** Probability and Statistics – teaching assistant (TA)

2019 – 2026 Weekly tutorials and problem-solving sessions for Calculus I–III, Basic Calculus (MATH 141), Calculus for Engineering and Science (MATH 145), Analytic Geometry (MATH 111), Fundamentals of Mathematics, and Probability and Statistics

## Teaching Expertise

**Subjects taught.** Single and multivariable calculus, analytic geometry, foundations of mathematics and proof, probability and statistics, and methods of mathematical economics.

**Explaining mathematics.** Seven years of teaching first- and second-year students in English. I explain ideas step by step with examples, and then move to formal definitions and proofs.

**Problem solving.** Weekly problem-solving sessions since 2019, built around worked examples, typical mistakes and exam-style questions.

**Course coordination.** TA coordinator of MATH 141 in Fall 2026–2027; part of exam preparation, proctoring and grading for core courses since 2019.

## Publications and Preprints

2026 — F. Temur, **H. B. Özcan**. Three-term difference relations do not detect uniform distribution. *Preprint*, 2026. [ResearchGate]

2026 — **H. B. Özcan**. The spectrum of $(\xi\alpha^n)$ can be uncountable. *Preprint*, 2026. [arXiv:2609.07714]

2025 — F. Temur, **H. B. Özcan**. The higher regularity of the discrete Hardy–Littlewood maximal function. *Preprint*, 2025. [arXiv:2504.13019]

2022 — H. Göral, **H. B. Özcan**, D. C. Sertbaş. The Green–Tao theorem and the infinitude of primes in domains. *The American Mathematical Monthly*, 2022. [DOI]

2020 — **H. B. Özcan**, S. Taşkın. Rings with few units and the infinitude of primes. *Hacettepe Journal of Mathematics and Statistics*, 2020. [DOI]

2020 — H. Göral, **H. B. Özcan**. Several novel proofs of the infinitude of primes. *The Mathematics Student*, 2020. [ResearchGate]

## Talks and Posters

Nov 2025 — On higher regularity of discrete Hardy–Littlewood maximal function. *2nd Atlantic Conference in Nonlinear PDEs & Harmonic Analysis*, Instituto Superior Técnico, Universidade de Lisboa. Poster.

Sep 2025 — The higher regularity of the Hardy–Littlewood maximal function. *İzmir Mathematics Day 7*, Ege University.

May 2025 — On the regularity of the discrete Hardy–Littlewood maximal function. *I. Workshop on Applied Mathematics and Statistics*, İzmir University of Economics.

Jan 2025 — Discretization approach to the weak boundedness of maximal convolution operators. Department Seminar, İzmir Institute of Technology.

Jul 2024 — Hardy–Littlewood maximal function. Department Seminar, FernUniversität in Hagen, Germany.

May 2022 — Hardy–Littlewood maximal function theorem. Department Seminar, İzmir Institute of Technology.

Nov 2021 — A bridge between additive combinatorics and the infinitude of prime numbers. Department Seminar, İzmir Institute of Technology.

Oct 2020 — Mathematicians will never stop to provide new proofs of the infinitude of primes. *İzmir Mathematics Days 3*, Dokuz Eylül University.

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| Dec 2019 | Prime ideal theorem on number fields. Department Seminar, Dokuz Eylül University. |
| Sep 2019 | Sayı cisimlerinde asal ideal teoremi (Prime ideal theorem on number fields). *32nd National Mathematics Symposium*, Ondokuz Mayıs University. |
| Sep 2019 | A special case of Fermat's last theorem. *Algebra Workshop*, Manisa Celal Bayar University. |
| Sep 2019 | Prime ideal theorem on number fields. *İzmir Mathematics Days 2*, Dokuz Eylül University. |
| Nov 2018 | Congruent number problem. Department Seminar, Dokuz Eylül University. |
| Jun 2018 | Krull–Schmidt–Remak–Azumaya theorem. *İzmir Mathematics Days 1*, Yaşar University. |

## Professional Service

- Referee for *The Mathematical Gazette*.
- Coordination of the department seminar series, İzmir Institute of Technology.
- Maintenance of the Department of Mathematics website, İzmir Institute of Technology.

## Skills and Languages

**Computing:** LaTeX (lecture notes, exams, slides), Python and R (computation and statistics), Manim (mathematical animation), GitHub, HTML.

**Languages:** Turkish (native), English (professional working proficiency), Russian (beginner).